

Question 1:

Find FIVE mistakes in the proof.

Each mistake should correspond to either

- A violation of the *Directions for Writing Proofs*, OR
- One of the listed *Common Mistakes*.

Theorem

If n is an odd integer, then $n^2 - 1$ is divisible by 8.

Proof:

Assume n is odd. This means $n = 2a + 1$ for any integer a .

We want to show $n^2 - 1$ is divisible by 8, so suppose it is divisible by 8.

Then

$$\begin{aligned}n^2 - 1 &= (2a + 1)^2 - 1 \\&= 4a^2 + 4a + 1 - 1 \\&= 4a^2 + 4a \\&= 4(a^2 + a)\end{aligned}$$

Since $a^2 + a$ is always even, it equals $2b$ where b is any integer.

So

$$n^2 - 1 = 4(2b) = 8b$$

Therefore the theorem is proven because we assumed the result and obtained the same result.

SOLUTION:

1. Misuse of the word 'any': It should say 'for some integer a ', not 'any integer a '.
2. Assuming what is to be proved: The proof says 'suppose it is divisible by 8', which is circular reasoning.

3. Jumping to a conclusion: It states ' $a^2 + a$ is always even' without justification.
4. Violates Writing Direction (clear statement + self-contained proof) : It does not clearly restate the theorem or properly set up the assumption in a formal way.
5. Confusion between what is known and what is still to be shown : The final sentence admits the result was assumed and then concluded again.

Question 2:

Prove the statement and give the counterexample if it is false. For all integers a , b and c if $a \mid (b + c)$ and $a \mid (b - c)$, then $a \mid 4b$

SOLUTION:

Proof:

Assume $a \mid (b + c)$ and $a \mid (b - c)$.

Then there exist integers k and m such that $b + c = ak$ and $b - c = am$.

Add the two equations: $(b + c) + (b - c) = ak + am$.

This gives $2b = a(k + m)$.

Multiply both sides by 2:

$$4b = a(2k + 2m).$$

Since $(2k + 2m)$ is an integer, $a \mid 4b$.

Therefore, the statement is TRUE.